CSI4130 - Assignment 4 Report
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## 1.0 Description of Program & Work Distribution

For Assignment 4 of CSI4130 – Computer Graphics, our team created Cosmo-Surf, an interactive space-themed flythrough featuring Marvel's Silver Surfer as the centerpiece. We implemented two required techniques:

- Sphere mapping with an HDR environment map for reflective lighting
- A custom particle system to generate a starfield with realistic stellar classifications (O-M class) and glow effects

We used RGBELoader and EquirectangularReflectionMapping to apply environment lighting, giving the Silver Surfer a metallic, comic-accurate look. Instead of a generic spaceship, we chose a more creative approach with the Surfer navigating past planets, ships, and Galactus.

We also implemented:

- Interactive controls (WASD + QE + mouse + boost)
- A custom start screen with animations and controls guide
- A dynamic trail effect following the Surfer for added visual polish

In terms of accomplishments from this project, we all significantly increased our expertise with Three JS, particularly when loading and utilizing models from Blender or SketchFab. We were unfamiliar with how to create a particle system, but through some online learning and documentation, we were able to develop an in-depth star system. Additionally, we also were not familiar with programming sphere mapping for lighting and how to use an environment map. We are proud to be able to look back and reflect on the growth in our programming expertise throughout this course.

Here is a table depicting the different responsibilities and work conducted, split by each individual group member.

| Group Member: | Work Done on Assignment 4: |
|---|---|
| Ali | - Wrote the Report<br>- Loaded in Silver Surfer Model and Created Initial Particle System<br>- Added in the Camera to be positioned behind the Silver Surfer model to create the Interactive Flythrough feel.<br>- Enhanced the movement of the Silver Surfer, adding in a speed boost, and movement in 3 dimensions.<br>- Added in a particle trail on the Silver Surfer. |
| Tolu | - Implemented planets and more to enhance the Flythrough experience<br>- Added in numerous light sources, using environment mapping.<br>- Added in Galactus |

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| | - Added in more planets, and nebulas |
| Eli | - Thoroughly enhanced Particle System, creating full fledged stars instead of just cubes<br>- Added a Sun asset and as a light source to brighten the scene. |
| Mustafa | - Created initial movement system for the model and added in ambient and position lights.<br>- Adjusted the Camera from the initial version. |

## 2.0 Guide to Usage

To utilize our program, you need the same dependencies and libraries as the Three JS Labs throughout this course and need Node Js to be installed. To launch and see our program, we have been utilizing `npx vite .` in the terminal to launch the HTML browser.

Upon launch, you will be greeted with the Silver Surfer standing and awaiting user input. You can control the Silver Surfer utilizing the W,A,S,D keys, with W representing forward movement, A representing leftward movement, S representing backward movement, and D representing rightward movement. Q and E are also used for movement, with Q representing an upward movement and E representing a downward movement.

Clicking the SHIFT key while moving will provide a 2x movement boost to speed stats. Beyond that, feel free to fly around and experience the space we have developed.

## 3.0 Acknowledgement of External Resources

In terms of external resources, we have used assets that were open source from SketchFab within our program. Below find a table depicting what was utilized and from where. We also made reference to some Stack Overflow pages and Three JS documentation, we have cited that below as well.

| Asset / Resource: | From Where It Was Obtained / Benefit: |
|---|---|
| Silver Surfer Model | https://sketchfab.com/3d-models/silver-surfer-5eff2b12bad14f1a8cc81fd7924d9a58 |
| Sun Model | https://sketchfab.com/3d-models/sun-9ef1c68fbb944147bcfcc891d3912645 |

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| Venus Planet Model | https://sketchfab.com/3d-models/venus-b306aaadbf2b4fcea1afa2db5ed75b4f |
| Destroyed Planet Model | https://sketchfab.com/3d-models/galactus-season-14-end-of-season-event-546c99e6eef24a3dadb1dad4ba6793c3 |
| Jedi StarFighter Model | https://sketchfab.com/3d-models/jedi-star-fighter-0b641c2f2b854f1f9ae7f2a731e44dbd |
| Galactus Model | https://sketchfab.com/3d-models/galactus-season-14-end-of-season-event-546c99e6eef24a3dadb1dad4ba6793c3 |
| https://discourse.threejs.org/t/where-to-find-realistic-hdr-space-textures-from-the-universe/43280 | Helped us develop an understanding of HDR. |